Adaptive Threat Recognition via Image Classification Using ResNet Architectures

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Abstract

Automated visual threat detection plays an increasingly important role in public-safety and surveillance systems. This work develops a lightweight, high-accuracy image classifier for distinguishing between threat and non-threat scenes using transfer learning with ResNet architectures implemented in FastAI. Our system uses a curated dataset containing firearms, knives, and challenging hard negatives such as phones, tools, and toy guns to reduce false positives. Leveraging a ResNet-34 model pretrained on ImageNet, combined with an optimized data pipeline and fine-tuning strategy, our classifier achieves roughly 92% accuracy with strong precision and recall for threat detection. We further analyze failure cases and discuss how dataset design, augmentation, and thresholding influence real-world deployment. Results demonstrate that efficient CNN architectures can provide effective threat-screening capabilities on modest datasets while maintaining computational practicality.

CCS Concepts

• **Computing methodologies** → **Object recognition**; • **Security and privacy** → *Surveillance.*

Keywords

threat detection, image classification, deep learning, ResNet, FastAI, computer vision

1 Introduction

Accurate and reliable threat recognition from visual imagery is critical for campus security, transportation hubs, and large public venues. Traditional computer-vision methods—largely based on hand-engineered descriptors such as SIFT and SURF—struggle in visually complex and cluttered environments due to limited generalization capacity [6]. Deep learning, particularly convolutional neural networks (CNNs), has demonstrated significant improvements across diverse classification tasks including facial expression recognition, defect detection, and workplace safety assessment [4, 5].

Building on these developments, our project investigates whether lightweight CNN architectures, specifically ResNet-based models fine-tuned using FastAI, can effectively distinguish between threat and non-threat images in realistic security scenarios. A primary challenge is minimizing false positives—misidentifying harmless objects as weapons—which can lead to unnecessary alarm escalation or operator fatigue. To mitigate this, we introduce hard negatives such as toy guns and metallic tools that visually resemble weapons.

Our objectives are to:

(1) Construct a curated dataset with balanced threat and non-threat classes, including hard-negative samples.
(2) Fine-tune a pretrained ResNet model using FastAI's training policies to maximize generalization.
(3) Evaluate the system using appropriate classification metrics and analyze performance limitations.
(4) Identify failure modes that inform future improvement strategies for real-world deployment.

We hypothesize that ResNet-based transfer learning is sufficient to achieve high threat-classification performance on modestly sized datasets when paired with robust data curation and augmentation.

2 Related Work

Research in threat detection and image classification reveals two converging directions: (1) leveraging FastAI for practical transfer learning workflows, and (2) applying CNNs to traditional safety-risk environments.

2.1 FastAI and Transfer Learning

FastAI has been shown to streamline deep learning experimentation while providing competitive performance even with limited datasets. Sitompul et al. used ResNet-50 for fabric defect classification and reported high accuracy for binary classes [1]. Zhang et al. improved ResNet-34 classification performance for garbage image classification using one-cycle learning and data augmentation [3]. These findings highlight FastAI's effectiveness for rapid prototyping and efficient fine-tuning.

2.2 Human Activity and Expression Recognition

Parvathaneni et al. applied CNNs and FastAI to human activity recognition, demonstrating that the framework generalizes well across different domains [2]. Similarly, Srinivasu et al. used deep CNN architectures within FastAI for facial expression analysis, showing that reliable performance can be achieved even with constrained datasets [4].

2.3 Threat and Safety Classification

Tepe and Eti examined the use of pretrained ResNet models for workplace risk-level prediction, showing promising results despite challenges from small and imbalanced datasets [5]. Mathew and Aziz explored traditional SIFT+SURF pipelines for X-ray threat detection, reaching nearly 96% accuracy [6], but such methods remain limited in generalization when applied to heterogeneous, natural imagery.

2.4 Real-Time Vision Architectures

Narsina et al. surveyed neural architectures for real-time image and video processing, emphasizing lightweight models such as MobileNet, YOLO, and Vision Transformers (ViT) optimized for edge deployment [7]. Their conclusions highlight that compact CNNs like ResNet-18/34 offer a strong trade-off between performance and efficiency, making them suitable for threat analysis under resource constraints.

Finally, Lu and Weng reviewed general image-classification techniques and strategies for improving classification performance, including balanced sampling and robust validation methodology [8]. Many of these principles remain relevant for modern CNN-based systems and inform our dataset design and evaluation.

3 Data

Our custom dataset contains two classes:

- **Threat**: images containing firearms, knives, and other weapon-like objects.
- **Non-threat**: images containing benign objects such as phones, tools, backpacks, and toy guns, including items that may resemble weapons in silhouette or texture.

3.1 Data Collection

Images were collected automatically via a Google Colab pipeline querying an image search provider (DuckDuckGo) using the `ddgs` library. A multi-query expansion strategy generated diverse search variations from base prompts such as "person holding knife in hand," including modifiers like "closeup," "side view," "indoors," "outdoors," and varying subject descriptors. All image URLs were globally deduplicated, and downloads that failed (broken links, unsupported formats, timeouts) were skipped.

The downloaded images were organized into class-specific directories (`threat/` and `non_threat/`) to support reproducible loading and training. After initial collection, we performed dataset cleaning to remove corrupted files, non-image content, and obvious mislabels.

3.2 Hard Negatives

To minimize false positives, we intentionally included hard negatives, such as:

- toy guns,
- hand tools (e.g., wrenches, screwdrivers),
- smartphones and other dark rectangular objects,
- everyday items with weapon-like shapes or textures.

These samples encourage the model to learn discriminative features and contextual cues rather than relying solely on superficial shape priors.

3.3 Preprocessing and Splits

For all experiments, images were:

- resized to 224×224 pixels,
- augmented with horizontal flips, small rotations (up to 10°), zoom, slight warping, and lighting variation,
- normalized using ImageNet mean and standard deviation.

We maintained near-equal representation of threat and non-threat classes, using targeted collection and, if necessary, augmentation to reduce class imbalance. The dataset was split into training (80%), validation (10%), and test (10%) subsets. All loading and transformation logic was implemented using FastAI's `DataBlock` API for clarity and reproducibility.

4 Methods

To address the challenge of automated threat detection in static images, we employed a transfer-learning approach using the FastAI library built on top of PyTorch. This allowed us to leverage state-of-the-art deep CNN architectures without requiring the massive datasets typically needed for training from scratch.

4.1 Model Architecture

We utilized a ResNet-34 architecture pretrained on the ImageNet dataset. ResNet (Residual Network) employs skip connections to allow gradients to flow more easily during backpropagation, mitigating the vanishing gradient problem in deeper networks. We replaced the original classification head with a custom fully connected layer outputting logits for our binary classification task ("threat" vs. "non-threat").

4.2 Data Pipeline and Preprocessing

Data ingestion was implemented via FastAI's `DataBlock` API with a random splitter (seed = 42) to ensure reproducible train/validation splits. The transformation pipeline consisted of:

(1) **Resizing**: images resized to 224×224 pixels, with padding in reflection mode to preserve aspect ratio without distortion.
(2) **Augmentation**: FastAI's `aug_transforms` were used to apply horizontal flips, rotations (up to 10°), zoom, perspective warping, and lighting variations.
(3) **Normalization**: input pixels normalized using ImageNet statistics to align the distribution with pretrained weights.

A representative grid of images (labeled as threat/non-threat in our dataset/pipeline) is shown below.

Figure 1: Representative image grid used during data inspection, showing examples from both classes (threat vs. non-threat).

4.3 Training Strategy

Training was conducted in two phases using FastAI's `fit_one_cycle` policy, which varies the learning rate and momentum to achieve faster convergence and better generalization:

(1) **Frozen Training (Phase 1)**: the convolutional backbone was frozen, and only the new head was trained for 3 epochs with a base learning rate of 2×10^{-3}. This allowed the classifier to adapt to the dataset without disrupting pretrained features.
(2) **Fine-Tuning (Phase 2)**: the entire model was then unfrozen. We used the learning-rate finder (`lr_find`) to identify an optimal learning-rate range and applied discriminative learning rates across layers. Fine-tuning continued for 10 epochs using the Adam optimizer with cross-entropy loss.

4.4 Evaluation Metrics

We evaluated the model using:

- accuracy,
- precision,
- recall,
- F1-score,
- confusion matrix,
- top-loss inspection (images where the model was confident but incorrect).

For security applications, precision and recall for the threat class are particularly important: precision controls false alarms, while recall indicates how many true threats are detected.

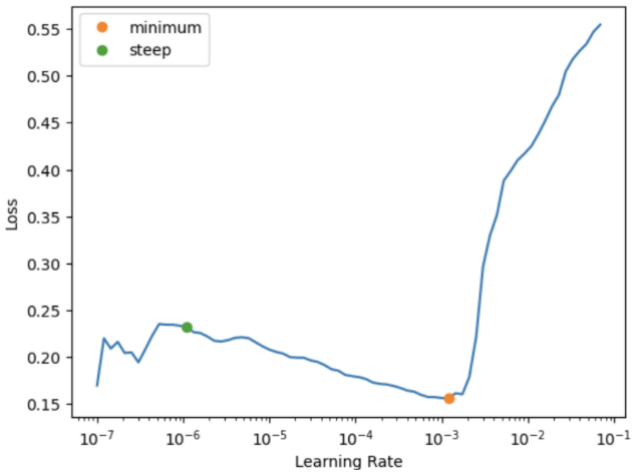

Figure 2: Learning rate vs. loss curve from FastAI's learning-rate finder. The selected learning-rate region corresponds to the steepest loss decrease before instability.

5 Results

5.1 Training Performance

During the initial frozen phase, the model quickly adapted to the new task, achieving a validation accuracy of approximately 92.7% by epoch 2 with validation loss around 0.25. After unfreezing and fine-tuning, the validation loss stabilized near 0.26, and the model reached a peak validation accuracy of roughly 92.4%. These results suggest that the ResNet-34 backbone successfully extracted relevant features for threat detection even with a modest dataset size.

5.2 Quantitative Evaluation

Final evaluation was conducted on the validation set containing 344 images. Table 1 summarizes the classification performance.

The model achieved an overall accuracy of 92%. Notably, the precision for identifying threats was 0.92, indicating a low false-positive rate, which is crucial for minimizing false alarms in security scenarios. The recall for threats was 0.89, meaning the system successfully identified 89% of all weapon-related images in the validation set.

5.3 Confusion Matrix and Error Modes

We examined a confusion matrix to better understand misclassifications.

Table 1: Classification report on the validation set.

Class	Precision	Recall	F1-score	Support
Non-threat	0.92	0.94	0.93	197
Threat	0.92	0.89	0.91	147
Overall accuracy		0.92 (344 images)		

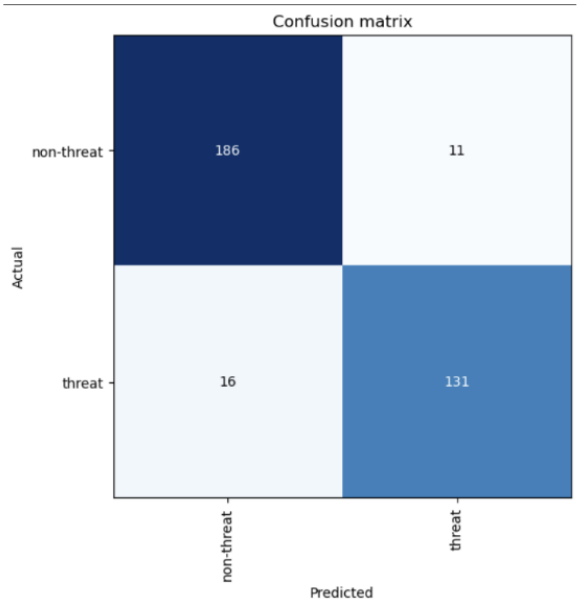

Figure 3: Confusion matrix for the binary threat vs. non-threat classification task on the validation set.

The model demonstrated a slight tendency to misclassify actual threats as non-threats (false negatives), as indicated by the somewhat lower recall for the threat class compared to non-threats. Top-loss inspection revealed that many difficult cases involved:

- occlusions (e.g., hands partially covering weapon handles),
- low-light or noisy conditions,
- atypical or extreme viewing angles,
- small weapon regions relative to a large background.

These error modes suggest that performance is currently constrained more by data diversity and edge-case coverage than by model capacity.

6 Discussion

Our results show that lightweight ResNet architectures paired with a carefully curated dataset can achieve high-accuracy threat classification. The strong precision for the threat class demonstrates robustness against false alarms—essential for real-world applications in which excessive false positives can erode operator trust or overload security staff.

However, the lower recall for threats indicates that the model occasionally misses subtle or challenging weapon instances. From a safety perspective, false negatives can be more critical than false positives. This trade-off emphasizes the importance of:

- expanding dataset diversity, particularly under challenging conditions (occlusion, low light, motion blur, partial visibility),
- incorporating targeted augmentation that explicitly simulates these failure modes,
- adjusting decision thresholds or applying calibration to favor recall in safety-critical deployments.

These insights align with prior work indicating that safety-critical classification is often bottlenecked by data quality and coverage rather than model architecture alone [8].

In addition, hard negatives appear to play a key role in reducing false positives: including toy guns, tools, and visually similar benign objects encourages the model to learn nuanced features and context rather than simplistic shape-based heuristics.

7 Conclusion

This work demonstrates that transfer learning with FastAI and a ResNet-34 backbone provides an effective and computationally efficient approach to binary threat detection in static images. Using a curated dataset containing both weapons and diverse non-threat hard negatives, our model achieved approximately 92% accuracy, with threat precision around 0.92 and threat recall around 0.89. These results validate our hypothesis that modest datasets, when carefully designed and augmented, can support robust performance for threat recognition.

Future work will focus on:

- expanding the dataset to better cover edge cases (night scenes, heavy occlusion, unusual poses),
- adding augmentation strategies that explicitly mimic real-world degradation (motion blur, sensor noise),
- exploring threshold-tuning or cost-sensitive learning to prioritize safety-critical behavior,
- comparing ResNet-34 against even lighter architectures (e.g., MobileNet, EfficientNet) for deployment on edge devices.

Overall, our findings suggest that ResNet-based transfer learning provides a strong foundation for practical, lightweight threat-detection systems in surveillance-style environments.

Figure 4: Example model predictions with confidence scores for threat and non-threat samples, illustrating correct classifications on representative cases.

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References

[1] E. Sitompul et al. Image classification of fabric defects using ResNet50 deep transfer learning in FastAI. *Bulletin of Electrical Engineering and Informatics*, 2024.
[2] N. S. Parvathaneni et al. Human Activity Recognition with CNN in FastAI. In *SASI-ITE Conference Proceedings*, 2024.
[3] Y. Zhang et al. Research on garbage image classification method based on ResNet-34 deep network. *International Journal of Applied Mathematics in Control Engineering*, 2021.
[4] P. N. Srinivasu et al. Deep Convolution Neural Network Model for Facial Expression Analysis Using FastAI. In *IEEE EICEEAI Conference Proceedings*, 2023.
[5] S. Tepe and S. Eti. A research on determining the degree of risk by using ResNet. *EPSTEM Proceedings*, 2023.
[6] L. S. Mathew and A. A. Aziz. Threat Detection Using Computer Vision Strategies. *International Journal of Industrial and Intelligent Engineering*, 2020.
[7] D. Narsina et al. Neural Network Architectures for Real-Time Image and Video Processing Applications. *Engineering International*, 2022.
[8] D. Lu and Q. Weng. A survey of image classification methods and techniques for improving classification performance. *International Journal of Remote Sensing*, 2007.